# Yoonho Lee

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**PROFESSIONAL SUMMARY**

**RESEARCH INTEREST**

**AI for Nucleic Acid Therapeutics**

- Virtual Cell, Drug Delivery System, Drug design, RNA/DNA therapeutics
- Specialized in Graph Neural Networks, Geometric Deep Learning and Categorical Deep Learning

**EDUCATION**

**KAIST (Korea Advanced Institute of Technology)**, Daejeon, South Korea

- Ph.D. in Graduate School of Data Science — Mar 2025 – Present
  - Research Interest: Virtual Cell, Drug Delivery System, Drug Design
  - Adviser: Prof. Chanyoung Park
- M.S. in Graduate School of AI — Mar 2023 – Feb 2025
  - Research Interest: Graph neural network, categorical deep learning, category theory
  - Adviser: Prof. Chanyoung Park
- B.S. in School of Computing, Electrical Engineering (double major) — Mar 2017 – Feb 2023

**PUBLICATIONS**

( *: Equal contribution )

**CONFERENCES**

[C7] Screening Lipid Nanoparticles through Structure-Ratio Alignment
**Yoonho Lee**, Yunhak Oh, Hoyoung Choi, Chanyoung Park
**NeurIPS 2026** - The Fortieth Annual Conference on Neural Information Processing Systems

[C6] scTrilemma: Balancing Identity, Invariance, and Fidelity in Single-Cell Representation Learning
Yunhak Oh, **Yoonho Lee**, Junseok Lee, Namkyeong Lee, Sangyeon Hwang, Yinhua Piao, Hyomin Kim, Seonghwan Kim, Jaechang Lim, Wooyoun Kim, Sungsoo Ahn, Chanyoung Park
**NeurIPS 2026** - The Fortieth Annual Conference on Neural Information Processing Systems

[C5] Disentangling Hyperedges through the Lens of Category Theory
**Yoonho Lee**, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, Chanyoung Park
**NeurIPS 2025** - The Thirty-Ninth Annual Conference on Neural Information Processing Systems

[C4] Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction
Sangwoo Seo, Jimin Seo, **Yoonho Lee**, Donghyeon Kim, Hyejin Shin, Banghyun Sung, Chanyoung Park
**ICCAD 2025** - International Conference on Computer-Aided Design

[C3] Subgraph Federated Learning for Local Generalization
Sungwon Kim, **Yoonho Lee**, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, Chanyoung Park
**ICLR 2025 (Oral, top 1.8%)** - The Thirteenth International Conference on Learning Representations

[C2] Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, Jihyeong Jung, **Yoonho Lee**, Chanyoung Park
**KDD 2024** - ACM SIGKDD Conference on Knowledge Discovery and Data Mining

[C1] IVIST: interactive video search tool in VBS 2021
**Yoonho Lee***, Heeju Choi*, Sungjune Park, YongMan Ro
**MMM 2021** - International Conference on Multimedia Modeling

**WORKSHOPS**

[W3] scTrilemma: Balancing Identity, Invariance, and Expression in Single-Cell Representation Learning
Yunhak Oh, **Yoonho Lee**, Junseok Lee, Namkyeong Lee, Sangyeon Hwang, Yinhua Piao, Hyomin Kim, Seonghwan Kim, Jaechang Lim, Wooyoun Kim, Sungsoo Ahn, Chanyoung Park
**NeurIPS 2026** - The Third Workshop on GenAI for Health: Agentic Systems, Clinical Trust, and Future Potential

[W2] Capturing functional context of genetic pathways through hyperedge disentanglement
**Yoonho Lee**, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, Chanyoung Park
**ICLR 2025** - Machine Learning for Genomics Explorations

[W1] Subgraph Federated Learning for Local Generalization
Sungwon Kim, **Yoonho Lee**, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, Chanyoung Park
**KDD 2024 (Best Paper Award)** - Workshop on Federated Learning for Data Mining and Graph Analytics

**PROJECTS**

- Circuit Recognition for Similar Circuit Search in Design Database, SK hynix 2024
- Development of a Scalable AI Reasoning Model for Knowledge System Expansion and Biomedical Applications 2025 –

**HONORS & AWARDS**

- **Best Paper Award** 2024
  KDD 2024 - Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), Barcelona, Spain
- **KAIST IE Frontier** 2021
  Third prize - topic: Development and Expansion of a Multimodal Data-Based Product Category Classification System
- **Reviewer**
  Top Reviewer - [NeurIPS] The Fortieth Annual Conference on Neural Information Processing Systems 2026

**PROFESSIONAL SERVICES**

**Organizer**

- Workshop
  - [MM4SciReal] Multimodal AI for Science and the Real World (ACCV workshop) 2026

**Conference Reviews**

- Program Committee
  - [AAAI] AAAI Conference on Artificial Intelligence 2026 – 2027
  - [CIKM] ACM International Conference on Information and Knowledge Management 2025 – 2026
- Reviewer
  - [NeurIPS] Conference on Neural Information Processing Systems 2025 – 2026
  - [ICLR] International Conference on Learning Representations 2025 – 2027
  - [ICML] International Conference on Machine Learning 2026
  - [LOG] Learning on Graphs Conference 2025
  - [KDD] ACM SIGKDD Conference on Knowledge Discovery and Data Mining 2024, 2026

**Workshop Reviews**

- Reviewer (NeurIPS, ICML, ICLR workshops)
  - [MLGenX] Machine Learning for Genomics Explorations 2025– 2026
  - [GenBio] Generative AI and Biology 2025
  - [AI4D3] AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development 2025
  - [AI4Science] AI for Science: The Reach and Limits of AI for Scientific Discovery 2025– 2026
  - [AIDrugX] AI for New Drug Modalities 2024
  - [Gen$^2$] Generative AI in Genomics (Gen2): Barriers and Frontiers 2026

**TEACHING EXPERIENCE**

- **DS535: Recommender System and Graph Machine Learning**, KAIST 2024
  Teaching Assistant

**TALKS AND SEMINARS**

- **Top Conference Session**, Korea Software Congress (KSC)
  Disentangling Hyperedges through the Lens of Category Theory 2025
- **Top Conference Session**, Korea Software Congress (KSC)
  Screening Lipid Nanoparticles through Structure-Ratio Alignment 2026

**REFERENCES**

- **Prof. Chanyoung Park**, Assistant Professor, KAIST
  Email: cy.park@kaist.ac.kr

[*CV compiled on 2026-11-05*]